

WASP-50b

R-1408

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_191130 · created 2026-08-01T04:53:51+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458817.786 +/- 0.022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.05
Transit depth (Rp/Rs)^2	3.05 +/- 1.76 [%]
Orbital Inclination (inc)	89.72 +/- 2.91
Airmass coefficient 1 (a1)	9.51 +/- 5.29
Airmass coefficient 2 (a2)	-1.22 +/- 0.81
Scatter in the residuals of the lightcurve fit is	56.92 %
Best Comparison Star	None
Optimal Aperture	2.53
Optimal Annulus	7.0
Transit Duration (day)	0.079 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.05 vs 0.1404 (deviation 0.49σ)
Duration fit vs published	0.079 d vs 0.0754 d (deviation 0.17σ)
Mid-time O–C	48.2 min
Depth SNR	2.5
Residual scatter	56.92 %

Light curve

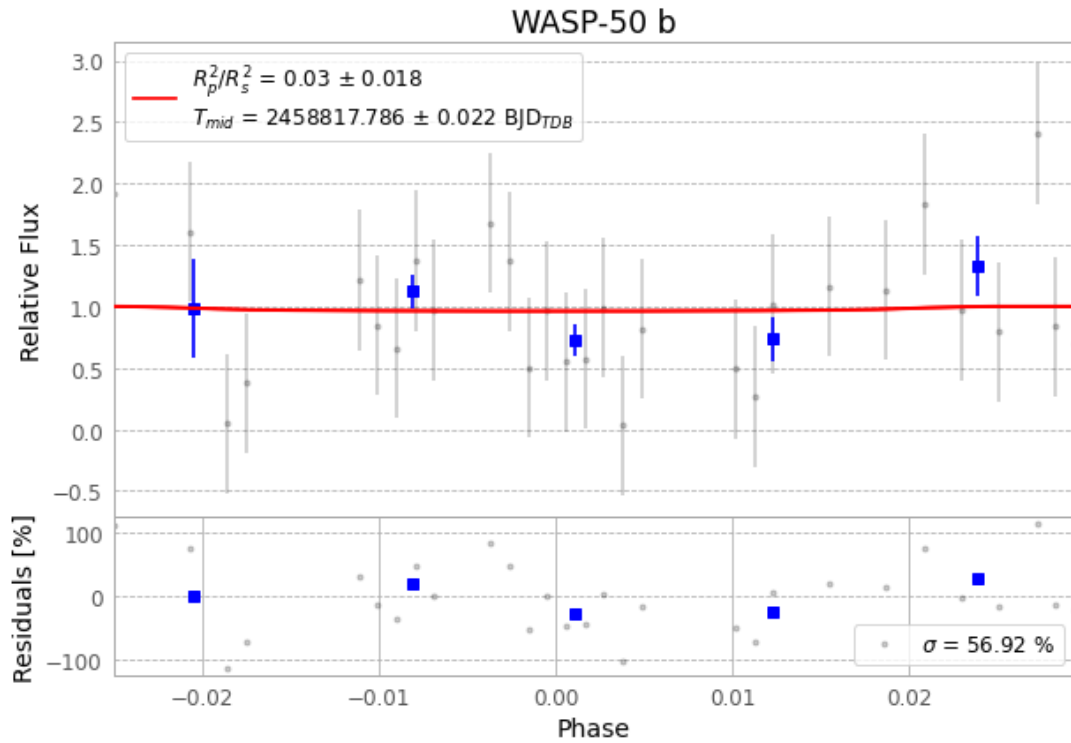


Figure 1 — FinalLightCurve_WASP-50 b_2019-11-29.png (SHA-256 e1b51f533fabe797...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [350, 231], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3bd994b935a60eb1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

54 FITS frames (35 MB), WASP-50191130052712.FITS → WASP-50191130080612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-191130.log (18bc5ab99d20...), FinalLightCurve_WASP-50 b_2019-11-29.png (e1b51f533fab...), FinalLightCurve_WASP-50 b_2019-11-29.csv (3e4b81f07a41...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 04:53Z.